

Modeling and control of cyber-physical systems (CPSs)

Final project proposal (Part II): formation control of nonlinear planar mobile robots

Diego Regruto, Sophie Fosson



POLITECNICO
DI TORINO

DAUIN

Department of Control and Computer Engineering

Academic year 2025–2026

Project goal

Nonlinear agents, output feedback linearization, formation control

In this final project, you will consider a team of three heterogeneous planar mobile robots.

- Each robot moves on the plane and has nonlinear affine dynamics.
- The measured and controlled output is the planar position

$$y_i = \begin{bmatrix} p_{x,i} \\ p_{y,i} \end{bmatrix}.$$

- First, you must apply input-output feedback linearization.
- Then, on the linearized input-output model, you must design a distributed formation controller.

After feedback linearization, each agent must be described by

$$\ddot{y}_i = \nu_i, \quad \nu_i = \begin{bmatrix} \nu_{x,i} \\ \nu_{y,i} \end{bmatrix}.$$

Agent structure

State, input, and output

For each agent $i = 1, 2, 3$, the state is

$$x_i = \text{col}(p_{x,i}, v_{x,i}, p_{y,i}, v_{y,i}) \in \mathbb{R}^4,$$

the physical input is

$$u_i = \begin{bmatrix} u_{x,i} \\ u_{y,i} \end{bmatrix} \in \mathbb{R}^2,$$

and the measured output is

$$y_i = \begin{bmatrix} p_{x,i} \\ p_{y,i} \end{bmatrix} \in \mathbb{R}^2.$$

The models are written in the common form

$$\dot{p}_i = v_i, \quad \dot{v}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i.$$

The virtual input generated by the formation controller is denoted by ν_i .

Input-output feedback linearization

What you have to verify

For each agent, prove that

$$\dot{y}_i = v_i,$$

and

$$\ddot{y}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i.$$

Then check that the decoupling matrix is nonsingular:

$$\det B_i(p_i, v_i) \neq 0.$$

If this holds, the input-output feedback linearizing law is

$$u_i = B_i(p_i, v_i)^{-1} [\nu_i - \phi_i(p_i, v_i)].$$

With this choice,

$$\ddot{y}_i = \nu_i.$$

Relative degree

Why there are no zero dynamics

For all three agents, the output is

$$y_i = \begin{bmatrix} p_{x,i} \\ p_{y,i} \end{bmatrix}.$$

The first derivative is

$$\dot{y}_i = \begin{bmatrix} v_{x,i} \\ v_{y,i} \end{bmatrix},$$

where the input does not appear. The second derivative is

$$\ddot{y}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i,$$

where the input appears through the decoupling matrix B_i . Thus the vector relative degree is

$$(r_x, r_y) = (2, 2), \quad r_x + r_y = 4.$$

Since $\dim(x_i) = 4$, the input-output feedback linearization linearizes the complete dynamics and there are no zero dynamics.

Nonlinear agents

What is given and what must be derived

For each of the three agents, you are given the nonlinear model in the form

$$\dot{p}_i = v_i, \quad \dot{v}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i.$$

The output is always

$$y_i = \begin{bmatrix} p_{x,i} \\ p_{y,i} \end{bmatrix}.$$

For each agent, you must:

- 1 compute $\dot{y}_i$ and $\ddot{y}_i$;
- 2 identify $\phi_i(p_i, v_i)$ and $B_i(p_i, v_i)$;
- 3 verify that $B_i(p_i, v_i)$ is nonsingular;
- 4 derive the feedback-linearizing law that yields $\ddot{y}_i = v_i$.

Agent 1

Planar robot with anisotropic nonlinear drag

The first agent is

$$\dot{p}_x = v_x,$$

$$\dot{v}_x = -0.35v_x - 0.08v_x|v_x| + \frac{1}{1.2}u_x,$$

$$\dot{p}_y = v_y,$$

$$\dot{v}_y = -0.45v_y - 0.10v_y|v_y| + \frac{1}{1.2}u_y.$$

For this agent, carry out the input-output feedback-linearization steps listed before.

- Verify the vector relative degree.
- Compute the decoupling matrix.
- Derive the physical input u_1 as a function of the virtual input ν_1 .

Agent 2

Planar robot with position-dependent apparent mass

The second agent is

$$\dot{p}_x = v_x,$$

$$\dot{v}_x = -0.40v_x + \frac{1}{1 + 0.25 \sin^2(p_x)} u_x,$$

$$\dot{p}_y = v_y,$$

$$\dot{v}_y = -0.55v_y + \frac{1}{1 + 0.30 \cos^2(p_y)} u_y.$$

For this agent, carry out the input-output feedback-linearization steps listed before.

- Verify the vector relative degree.
- Compute the decoupling matrix.
- Derive the physical input u_2 as a function of the virtual input v_2 .

Agent 3

Planar robot with nonlinear coupling

The third agent is

$$\dot{p}_x = v_x,$$

$$\dot{v}_x = -0.30v_x + 0.15 \sin(p_y)v_y + (1 + 0.20 \cos^2(p_y)) u_x + 0.08 \sin(p_x)u_y,$$

$$\dot{p}_y = v_y,$$

$$\dot{v}_y = -0.35v_y + 0.15 \cos(p_x)v_x + 0.08 \sin(p_y)u_x + (1 + 0.20 \sin^2(p_x)) u_y.$$

For this agent, carry out the input-output feedback-linearization steps listed before.

- Verify the vector relative degree.
- Identify the drift term and the decoupling matrix.
- Prove that the decoupling matrix is nonsingular.
- Derive the physical input u_3 as a function of the virtual input v_3 .

After feedback linearization

Common double-integrator model

After the local input-output feedback linearization, all agents satisfy

$$\ddot{y}_i = \nu_i, \quad i = 1, 2, 3.$$

Equivalently,

$$\dot{y}_i = w_i, \quad \dot{w}_i = \nu_i.$$

The multi-agent formation controller must be designed at the level of the virtual input ν_i . The physical input applied to the nonlinear robot is then recovered locally from

$$u_i = B_i(p_i, v_i)^{-1} [\nu_i - \phi_i(p_i, v_i)].$$

Suggested formation task

Triangular moving formation

Use the three agents to achieve a triangular formation. Let

$$h_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad h_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad h_3 = \begin{bmatrix} 0.5 \\ \sqrt{3}/2 \end{bmatrix}.$$

The desired moving formation is

$$y_i^*(t) = r(t) + h_i - h_1.$$

Possible references are:

- constant velocity motion;
- circular motion;
- sinusoidal motion.

The output formation error relative to node 1 is

$$\delta_i(t) = (y_i(t) - y_1(t)) - (h_i - h_1), \quad i = 2, 3.$$

Distributed-control options

Choose one design route

Each group may choose one of the following approaches.

- **State-feedback protocol** Design a distributed controller for the double-integrator model using state feedback on relative positions and velocities.
- **Loop-shaping protocol** Design a distributed output-feedback controller

$$\nu_i(s) = K(s)\varepsilon_i(s),$$

where $K(s)$ is chosen in the frequency domain.

In both cases, the controller must act on the virtual input ν_i , while the physical input u_i is obtained by local feedback linearization.

Required deliverables

What must be included

Each group must submit a short report and MATLAB/Simulink code including:

- ① derivation of ϕ_i and B_i for all three agents;
- ② verification of the vector relative degree $(2, 2)$;
- ③ computation of the feedback-linearizing law;
- ④ design of the distributed formation controller;
- ⑤ simulation of the multi-agent closed-loop system;
- ⑥ plots of positions, formation errors, and physical inputs.
- ⑦ Analysis of the behavior of the controlled systems considering, e.g., the aspects considered in the next slide

Possible additional aspects to be analyzed

Additional points for discussion

Possible extensions include:

- Simulate the nonlinear dynamics (applying the physical commands u_i).
- compare different communication graphs;
- study the effect of actuator bounds on v_i and/or u_i ;
- add output disturbances on the measured positions;
- add sensor noise;
- compare state-feedback and loop-shaping approaches;
- discuss the effect of Q , R , c , structure of $K(s)$, observer gain;
- discuss robustness with respect to imperfect model knowledge in the feedback-linearizing law.
- ...

Good luck ;-)!